

ASSIGNMENT CLO-1 (FE SW)**SPRING SEMESTER 2025****DUE DATE: 05-05-2025****MAX.MARKS:05**

Question 1: Discuss the difference between Diamagnetism, Para-magnetism and Ferromagnetism and the effect of temperature on magnetization

DIFFERENCE BETWEEN DIAMAGNETISM, PARA-MAGNETISM AND FERROMAGNETISM:-**a) Diamagnetism:-**

Diamagnetism is the most basic form of magnetism, exhibited by all materials to some extent. It arises from the orbital motion of electrons in atoms, which produces tiny circulating currents and hence magnetic moments. In diamagnetic substances, all electrons are paired, and there is no net permanent magnetic moment per atom. When an external magnetic field is applied, these induced magnetic moments align in a direction opposite to the applied field, resulting in a weak repulsive force. This effect is very weak and often masked by other stronger magnetic effects if they are present. Examples of diamagnetic materials include bismuth, copper, silver, gold, and quartz. Since the source of diamagnetism is purely quantum mechanical and does not involve thermal alignment, temperature has very little or no effect on the magnetic behaviour of diamagnetic materials.

b) Para-magnetism:-

Para-magnetism occurs in materials that have atoms or ions with unpaired electrons. These unpaired electrons possess a net magnetic moment. In the absence of an external magnetic field, these magnetic moments are randomly oriented due to thermal motion, resulting in no net magnetization. However, when an external magnetic field is applied, the magnetic moments tend to align with the field, resulting in a weak attraction. This alignment is only partial because it competes with thermal agitation. The magnetization is directly proportional to the strength of the applied field and inversely proportional to the temperature, as explained by Curie's Law. Common examples of paramagnetic materials are aluminum, platinum, and magnesium.

c) Ferromagnetism:-

Ferromagnetism is the strongest type of magnetism and is exhibited by materials like iron, cobalt, and nickel. In these substances, unpaired electrons in atoms interact with each other through quantum mechanical exchange forces, causing their magnetic moments to align spontaneously in the same direction within small regions known as domains. Even without an external magnetic field, the alignment of magnetic moments in these domains gives rise to a strong net magnetization. When an external field is applied, the domains grow in size and align further, significantly increasing the magnetization.

d)EFFECT OF TEMPERATURE ON MAGNETIZATION:-

- In **paramagnetic materials**, increasing the temperature increases the random thermal motion, which disturbs the alignment of magnetic moments, thereby reducing the net magnetization.
- In **ferromagnetic materials**, temperature plays a crucial role. As temperature increases, thermal agitation disrupts the alignment of domains. Above a certain critical temperature called the Curie Temperature, the ferromagnetic material loses its spontaneous magnetization and becomes paramagnetic.
- In **diamagnetic materials**, since the effect is not related to alignment of magnetic moments but rather induced currents, temperature has negligible effect.

Question 2: Compare the behaviour of conductor and insulator in an electric field.

a) Conductors

Conductors are materials that allow the free movement of electric charges, particularly electrons. When a conductor is placed in an external electric field, the free electrons inside it immediately start to move in response to the field. These electrons continue to move until they have rearranged themselves in such a way that they cancel the internal electric field. As a result, in electrostatic equilibrium, the net electric field inside a conductor is zero. Furthermore, excess electric charges in a conductor reside only on its outer surface, and the electric field just outside the surface is perpendicular to the surface. Conductors are essential in applications such as electrical wiring and shielding, where quick redistribution of charge is needed.

b)Insulators (Dielectrics)

Insulators, or dielectrics, are materials that do not have free charges available to move. The electrons in an insulator are tightly bound to atoms or molecules. When an insulator is placed in an electric field, the atomic charges do not move freely, but a small displacement of the positive and negative charges occurs, leading to polarization of the material. This polarization results in a reduced effective electric field within the material compared to the external field. Unlike conductors, insulators do not cancel the electric field inside them; rather, they allow it to penetrate, but to a lesser extent. This property is utilized in capacitors, where a dielectric material increases the capacitance by reducing the field strength between plates.

Question 3: As the projectile moves in its path is there any point along the path where the velocity and acceleration are (a) perpendicular to each other. (b) parallel to each other.

RELATIONSHIP BETWEEN VELOCITY AND ACCELERATION IN PROJECTILE MOTION**(a) When are velocity and acceleration perpendicular:-**

At the highest point of the projectile's path, the object is momentarily not moving upward or downward its vertical velocity becomes zero. However, it still moves horizontally, so it has a horizontal velocity. At that same point, the acceleration due to gravity is still acting vertically downward. Since one is horizontal and the other is vertical, they are at right angles (90°) to each other. So yes, there is a point (at the top of the path) where velocity and acceleration are perpendicular.

(b) When are velocity and acceleration parallel:-

Velocity and acceleration are parallel when they point in the same direction. This happens during the downward motion of the projectile that is, after the highest point, when the object is falling. In this case, the velocity is downward (since the object is moving down), and acceleration is also downward (because of gravity), so both are in the same direction. Hence, they are parallel during the downward motion. Similarly, in the upward motion, velocity is upward while acceleration is downward, so they are in opposite directions we say they are anti-parallel, not parallel.

Question 4: Discuss the damped harmonic oscillation and its types.

Damped Harmonic Oscillation:-

Damped harmonic oscillation refers to a type of motion where an object vibrates back and forth, but gradually loses energy over time due to resistance such as friction or air drag. As a result, the object's motion becomes slower, and the amplitude (height of vibration) keeps decreasing until the object eventually comes to rest. This type of motion is seen in many real-life systems where energy is not perfectly conserved.

Types of Damped Harmonic Oscillations

Damped oscillations are usually divided into three types based on how quickly the object comes to rest:

1. Under-Damped Oscillation

In an under-damped system, the object continues to oscillate (move back and forth), but each time the movement is weaker than before. The amplitude decreases slowly over time, but the object takes a while before stopping completely. This is the most common type of damping seen in everyday life. A good example is a pendulum that swings and gradually comes to a stop.

2. Critically Damped Oscillation

In a critically damped system, the object returns to its resting position very quickly without oscillating at all. It is the fastest possible return to rest without any back-and-forth motion. This type of damping is very useful in machines and systems where we want to avoid vibration and stop motion smoothly and quickly. For example, car shock absorbers are designed to be critically damped so that the car doesn't bounce after hitting a bump.

3. Over-Damped Oscillation

In over-damped systems, the object also does not oscillate, but it returns to its resting position very slowly. It takes longer to stop compared to a critically damped system. This happens when the damping force is too strong, and the system is heavily resisted. An example could be a door closer that shuts the door very slowly without bouncing.